

Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 17 papers, 2070–2081 BS

1. Introduction

2 hrs · in 16/17 papers · 2 topics, both TOP tier · no numericals in this ch.

Whole chapter in 2 items:

- KDD process **TOP 7/17**
 - DW fundamentals **TOP 9/17**
-

1.1 Data Mining Origin

What is DM? + Explain steps of KDD process 2+3 2+5 2+6 1+5 4

TOP 7/17 Most repeated Q in the paper.

DM

- Extraction of **interesting** patterns/knowledge from large data.
- Interesting = non-trivial + implicit + previously unknown + potentially useful.
- Not retrieval:
 - Query tool → answers what you already know to ask.
 - DM → finds patterns you didn't know to look for.
- Also called: KDD, knowledge extraction, data/pattern analysis, data archaeology, data dredging, BI.

Not DM

- Phone-book lookup, `SELECT AVG(salary)`, web search → info retrieval (query states exactly what is wanted).

Descriptive DM

- Characterise the general properties of data.
- Unsupervised.
- "What is in data? Unusual patterns?"
- Clustering, association, summarisation, visualisation.

Predictive DM

- Infer unknown $Y = f(X)$ from known X .
- Supervised.
- "Which customers leave in 6 months?"
- Classification, regression, prediction.

DM vs KDD

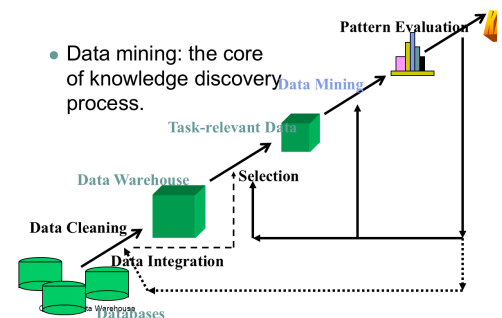
- Strictly: DM = **one step** of KDD (the step that runs the algorithms).
- In practice treated as synonyms. **IOE papers use them interchangeably.**
- **If Q says "steps of data mining" → write KDD steps.**

KDD: 7 steps (Han & Kamber) → default answer

1. **Data cleaning:** rm noise, smooth outliers, fill missing, fix inconsistency.
2. **Data integration:** combine multiple heterogeneous sources.
3. **Data selection:** retrieve only task-relevant data.
4. **Data transformation:** normalise, aggregate, generalise, discretise.
5. **Data mining:** apply algorithms. *Essential step.*
6. **Pattern evaluation:** keep truly interesting patterns (support, confidence, lift, novelty).
7. **Knowledge presentation:** visualise + represent for user.

Steps 1-4 = data preparation ≈ 60% of effort.

Data Mining: A KDD Process



Conflict in your sources verified/added

- Chapter 2_Data Warehouse.ppt → **6 stages** (Adriaans & Zantinge): selection → cleaning → enrichment → coding → mining → reporting.
- Han & Kamber (syllabus ref. book 2) → **7 steps** above.
- Both examined at IOE (old CT 725 paper asked "Cleaning, Enrichment, Coding").
- **Write 7-step version, add 1 line naming the 6-stage variant.**

Answer skeleton [5-7]

1. DM definition, 1 line + "one step of KDD".
2. **Draw the pipeline** (markers reward the fig).
3. 7 steps, **1 line each**. No more.
4. Close: "steps 1-4 = data prep ≈ 60% effort".

Origin of DM: why it emerged 2 3

Confluence of disciplines

- DB systems
- Statistics
- Machine learning
- AI / pattern recognition
- Visualisation
- High-performance computing

Drivers

- Automated collection → data explosion (POS scanners, txn logs, sensors, satellites, Web).
- Cheap storage → hoarding became rational. Analysis did not keep pace.
- Competition → unmined data = wasted asset.

Why traditional/manual analysis fails

- Enormity (TB scale).
- **High dimensionality** (1000s of attrs).
- Sources heterogeneous + distributed.
- Data collected for **ops**, not analysis → noisy, incomplete, inconsistent.

DM vs query tools (SQL)

- SQL: must know *exactly* what you want. DM: only *vaguely*.
- SQL → facts/aggregates. DM → patterns/trends.
- SQL: deterministic retrieval. DM: ML/statistical search.
- Hidden relations:
 - SQL cant trace them
 - DM finds them automatically

“The world is data rich but information poor”. Justify. 8

PIN 1/17 73 Shr Justify-in-own-words → needs structure, not definitions.

- 1. State paradox:** TBs stored, decisions still on intuition. Data $\neq$ info $\neq$ knowledge.
- 2. Data rich:** POS logs, bank txns, telecom call records, remote sensing, Web. Growth automated.
- 3. Info poor, why:**
 - Volume defeats manual inspection.
 - Data is operational → noisy, incomplete, inconsistent.
 - Scattered across heterogeneous systems.
 - Query tools answer only what is asked.
 - Shortage of analysts.
- 4. Consequence:** “data tombs” = archives written, never read. Decisions made w/o evidence already paid for.
- 5. Resolution:**
 - DW consolidates + cleans
 - DM extracts the patterns
 - Data → Info → Knowledge → Decision
- 6. Eg:** retail chain holds millions of receipts (rich) but cant say which products sell together (poor) until association analysis gives {nappies} $\Rightarrow$ {beer}.

Architecture of a typical DM system (+Fig) 2 4+2

78 Bh asked as "Explain DM process with diagram".

- **Data sources:** DB, DW, WWW, other repos.
- **DB/DW server:** fetches data relevant to the request.
- **Knowledge base:** domain knowledge guiding search + judging interestingness (concept hierarchies, user beliefs, constraints).
- **DM engine:** characterisation, association, classification, clustering, outlier, evolution analysis.
- **Pattern evaluation:** applies interestingness measures. *Best pushed into the engine* so dull patterns are never generated.
- **UI:** specify task, browse schemas/hierarchies, visualise results.

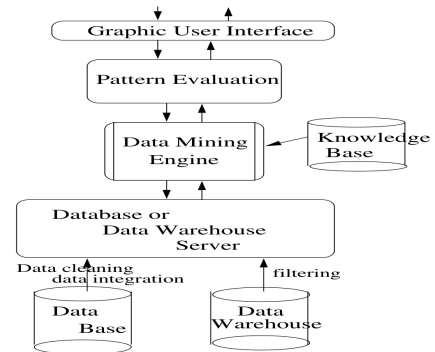


Figure 1.5: Architecture of a typical data mining system.

DM functionalities 2 4

- **Characterisation & discrimination:**
 - summarise the target class
 - contrast it against a comparative class
- **Association & correlation:** freq patterns, rules $X \Rightarrow Y$. → Ch 4
- **Classification & prediction:** assign class labels / predict numeric. → Ch 3
- **Cluster analysis:** group w/o pre-defined labels. → Ch 5
- **Outlier / anomaly:** objects not complying w/ general behaviour. Noise to discard, but *the prize* in fraud. → Ch 6
- **Evolution analysis:** trends for objects changing over time. → Ch 7

What is DM? Explain its applications. 10

PIN 1/17 71 Ch

Domain	What is mined
Biomedical / DNA	Similarity search across DNA seqs; association analysis for co-occurring genes (most diseases = gene combinations, not 1 gene); path analysis linking genes to disease stages.
Financial	DW design for multidim analysis; loan payment prediction, credit policy; classification + clustering for customer targeting; money-laundering detection .
Retail / market basket	Which products sell together; customer retention; campaign effectiveness; recommendation.
Telecom	Fraudulent usage patterns; multidim analysis of calls; mobile user behaviour.
Intrusion detection	Anomaly detection over traffic + audit logs. → Ch 6
Web mining	PageRank, usage + structure mining. → Ch 7

1.2 Data Mining & Data Warehousing Basics

What is a DW? Write its key features. 2 2+4 2+5

TOP 9/17

Definition (Inmon)

- DW = **subject-oriented, integrated, time-variant, non-volatile** collection of data supporting management decision-making.
- **Memorise as S-I-T-N** → this is the whole “key features” answer.
- **S = Subject-oriented**
 - Organised around subjects (customer, product, sales), not daily ops.
 - Data useless for decisions is excluded.
- **I = Integrated**
 - Built from heterogeneous sources (relational DBs, flat files, txn records).
 - Cleaning + integration force consistent naming, encoding, units.
 - → 1 name per entity.
- **T = Time-variant**
 - Long history (5–10 yrs).
 - Every key structure carries a time element.
 - Operational DB holds only the *current* value.
- **N = Non-volatile**
 - Physically separate. No operational update.
 - Only 2 ops: **initial load** + **access**.
 - → no txn processing, recovery or concurrency control needed.
- **Data warehousing** = the *process* of building + using a DW (extract, transform operational data, load into central store).

How is DW diff from a DB? How are they similar? 2+2

PIN 2/17 75 Ash Asks both directions.

	Operational DB (OLTP)	Data warehouse (OLAP)
Purpose	Run day-to-day business	Support analysis + decisions
Content	Current, detailed	Historical, summarised + detailed
Design	Highly normalised, many tables (ER)	De-normalised, few tables; star/snowflake
Access	Short read/write txns	Long, complex, read-mostly
Users	Clerks, DBAs, front-line	Managers, analysts, execs
Unit of work	Simple txn	Complex query
Records/query	Tens	Millions
Size	100 MB–GB	100 GB–TB

Similarities verified/added

- Both persistent, managed data stores.
- Both usually on a relational DBMS.
- Both use schemas, indexes, SQL.
- Both enforce integrity + security.
- DW is *built from* the DBs.

Why keep them separate

- **Performance:** opposite tuning.
 - DBMS → OLTP (indexing, concurrency, recovery)
 - DW → OLAP (multidim views, consolidation)
 - both on 1 system degrades both
- **Missing data:** decisions need history ops DBs dont keep.
- **Consolidation:** needs aggregation across heterogeneous sources.
- **Data quality:** inconsistent codes/formats reconciled once, centrally.

Explain DW architecture with its analytical processing 8

TOP 9/17 75 Ch 2 architectures in your sources: complementary, not contradictory (three-tier = layered, process = component-wise).

(a) Three-tier architecture (Han & Kamber) → standard answer

- **Bottom tier** = warehouse DB server (relational).
 - Data pulled via **gateways**: ODBC, OLE-DB, JDBC.
 - Back-end tools: extract, clean, transform, load, refresh.
 - Holds **metadata repo**.
- **Middle tier** = **OLAP server** (presents data as a cube).
 - **ROLAP**:
 - o extended relational DBMS
 - o maps multidim ops → relational ops
 - o scales to large data
 - **MOLAP**:
 - o special-purpose server, direct multidim arrays
 - o fast, but sparse data wastes space
 - **HOLAP**:
 - o hybrid of the two
 - o detail in relational, aggregates in arrays
- **Top tier** = front-end client: query & reporting, analysis, DM tools.

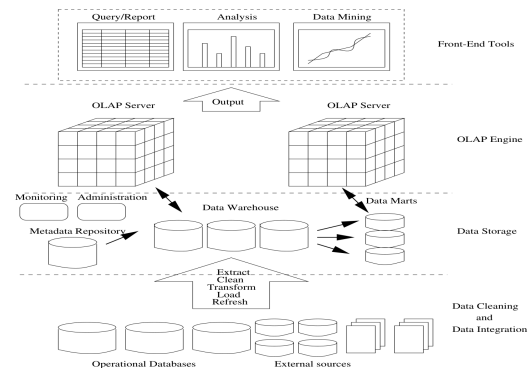
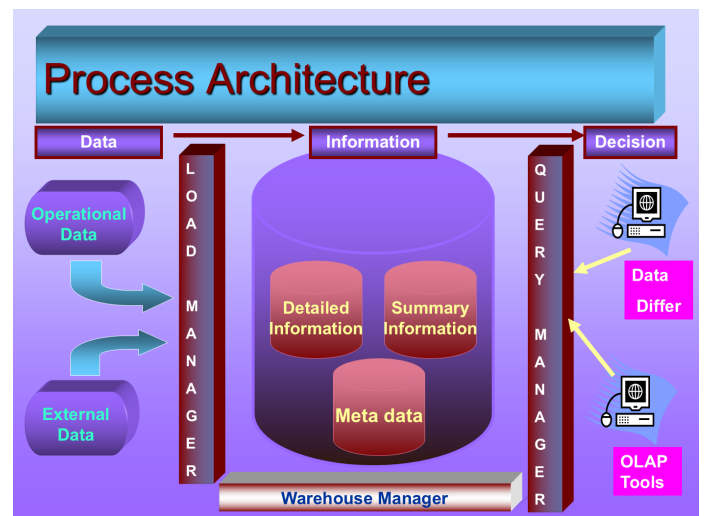


Figure 2.12: A three-tier data warehousing architecture.

(b) Process / component architecture (form drawn in class notes)

- **Load manager**: all extract + load ops. Fast-loads into temp store, simple transforms.
- **Warehouse manager**:
 - consistency + referential checks
 - merges temp store into the published DW
 - creates indexes + business views
 - updates aggregations
 - backup + archive
- **Query manager**:
 - directs queries to the right tables
 - stops 1 query starving the system
 - captures query profiles
- **Detailed info**: lowest-granularity data retained.
- **Summary info**:
 - pre-computed aggregations built by the warehouse manager
 - transient, changes as query profiles change
- **Metadata**: data about data. Describes how info is structured, maps sources to a common view.



(c) "...with its analytical processing" → 3 kinds of DW application

1. **Information processing**: querying, basic stats, reporting (crosstabs, tables, charts). *No new knowledge.*
2. **Analytical processing (OLAP)**: multidim analysis. Roll-up, drill-down, slice, dice, pivot. *Summarises, doesn't learn.*
3. **Data mining**: knowledge discovery from hidden patterns. *Creates new knowledge.*

What is DW and data mart? (short notes) 15 2+4

PIN 2/17 70 Ch, 71 Shr

Data mart

- Subset of a DW, confined to 1 user group / functional area, in its own DB.
- **Dependent:** sourced *from* an enterprise DW → consistent by construction.
- **Independent:** populated *directly* from sources, no central DW → faster to build, risks inconsistency between marts.
- **Benefit:**
 - less volume to scan → faster queries
 - each department owns its own structures
- **Eg:** departmental store sales mart, bank loan mart.

3 DW models

- **Enterprise warehouse:** all subjects, whole org. Cross-functional, large, slow to build.
- **Data mart:** subset for a specific group.
- **Virtual warehouse:**
 - set of views over operational DBs
 - only some summary views materialised
 - easy to build, but loads the operational systems

Describe Snowflake schema with example 4

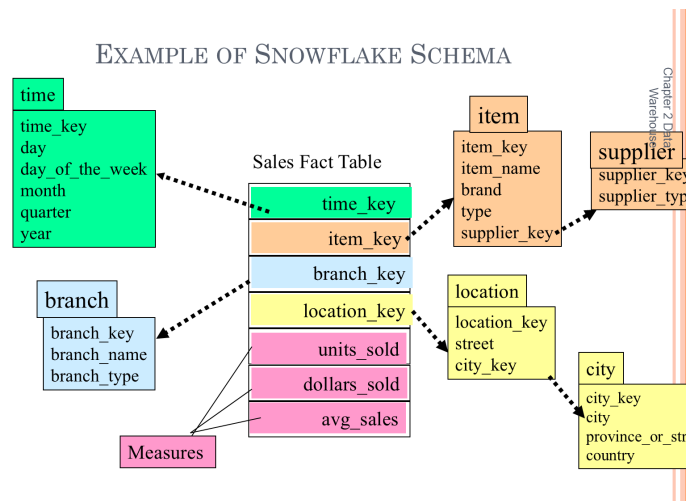
PIN 1/17 74 Ch

- **Star:** 1 central **fact table** (measures + FKs) + **de-normalised** dimension tables. Simple, fast joins.
- **Snowflake:** star refined, some dimension hierarchies **normalised** into smaller sub-dimension tables. Saves space, less redundancy, but more joins → slower.
- **Fact constellation (galaxy):** multiple fact tables sharing dimension tables = collection of stars.

Eg: sales DW

- FactSales: time_key, item_key, branch_key, location_key, units_sold, dollars_sold, avg_sales.
- location normalised: location → city → province.
- item → supplier.
- **That normalisation of the dimension hierarchy is what makes it snowflake, not star.**

→ Full multidim model (cuboids, base/apex, star-schema drawing numerical) sits in Ch 2, OLAP & Multidimensional Data Analysis, where it is examined harder.



Fundamental differences between DM and DW 3

PIN 2/17 76 Ash

	Data warehousing	Data mining
Nature	Storage architecture	Analysis process
Purpose	Consolidate + organise data	Discover hidden patterns in it
Output	Consistent query-able historical data	Rules, models, clusters, predictions
Done by	Engineers / DBAs	Analysts, statisticians, business users
Relation	DW = infrastructure; DM = 1 of its 3 uses. DW not required for DM, but a clean DW makes DM far more effective.	

How can a DM system be integrated with a DB/DW system? 4

PIN 1/17 78 Bh Answer = the 4 coupling schemes.

1. No coupling

- Uses no DB/DW function at all.
- Reads flat file → mines → writes to another file.
- Simple, but: poor data selection, no scalability, no consistency.

2. Loose coupling

- Mining system → fetches data from repo managed by DB/DW.
- Stores results back into a file / designated place in DB.
- Uses DBMS for storage + retrieval only.
- Cant exploit indexing or query optimisation → memory-bound, doesnt scale well.

3. Semi-tight coupling

- Besides linking, DB/DW provides efficient **mining primitives**.
- Primitives: sorting, indexing, aggregation, histogram analysis, multi-way join.
- + pre-computed stats: sum, count, max, min.
- → materially better performance.

4. Tight coupling

- DM system **smoothly integrated into** DB/DW as 1 functional component.
 - Mining queries optimised alongside DB queries.
 - Intermediate results can be stored.
 - Scales. **Most desirable design.**
-

1.3 PYQ Mapping

Tiers: **TOP /17** = 7+/17 papers **HOT /17** = 4-6 **PIN /17** = 1-3 but ≥ 8 marks. Same Q in diff wording grouped under 1 entry.

1.1 Data Mining Origin

1. **TOP 7/17** Define DM + explain KDD / DM process. 2+3 to 2+6
 - "What is data mining? Explain the process of data mining." 74 Ash
 - "...Explain all the steps of knowledge discovery." 72 Ch
 - "...What are the steps involved in knowledge discovery process?" 79 Bh
 - "...Explain the steps of KDD process briefly." 81 Ba
 - "...Explain general steps in brief." 72 Ka
 - "What is a Data Mining? Explain its application." 71 Ch → 2nd half swaps to applications.
2. **PIN 1/17** "The world is data rich but information poor". Justify. 8 73 Shr

1.2 DM & DW basics

3. **HOT 3/17** What is a DW / key features + architecture.
 - "What is data warehousing? Explain with example where data warehouses are used." 2+4 80 Ba
 - "Write key features of data warehouse. Explain each step of the KDD process w/ example." 2+5 80 Bh
 - "Explain Data Warehouse architecture with its analytical processing." 8 75 Ch
4. **PIN 2/17** DM vs DW / DW vs DB.
 - "What are the fundamental differences between Data Mining and Data Warehousing? Describe the steps of KDD." 3+7 76 Ash
 - "How is data warehouse different from a database? How are they similar?" 2+2 75 Ash
5. **PIN 1/17** "Explain how DM system can be integrated with DB/DW system. Explain DM process with diagram." 4+2 78 Bh
6. **PIN 1/17** "What is data warehouse and data mart? Describe Snowflake scheme with example." 2+4 74 Ch
7. **PIN 2/17** Short notes: **Data warehouse and Data mart.** 15 70 Ch, 71 Shr

Coverage check

- All Ch 1 PYQs covered above.
- Filled by research, absent from your notes verified/added :
 - *Similarity* half of "how are DW and DB similar?" 75 Ash (notes only contrast them).
 - Reconciliation of the **6-stage vs 7-step KDD** clash between your 2 sources.

Sources used for this chapter

- **Han & Kamber 3rd ed** (full textbook PDF): ch 1 KDD process, DM system architecture, functionalities, applications; ch 4 DW definition, 3-tier architecture, DW models, DW usage. **Primary authority; all of the above re-verified against it.**
- **Han & Kamber 1st ed manuscript** (ch01.pdf, ch2.pdf): source of the two architecture figures, and of the **4 coupling schemes**, which the 3rd ed drops. **78 Bh still examines coupling**, so it is kept here.
- Chapter 2_Data Warehouse.ppt, Chapter 4 and 5 DW Architecture.ppt, applicationDM.ppt, data mining_/Chapter 1_2.pdf.
- **No DBK lecture exists for Unit-1**, so class slides could not be used here.

Numericals

- **Ch 1 has no numerical component.** No Q in 17 papers needs a calculation here → no "Ch1 Numerical" companion. First numericals appear in Ch 2 (normalisation, similarity measures, PCA).

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.